

Q. Write a C program to generate truth tables for the following logical expressions:

1. p AND q (Conjunction)
2. p OR q (Disjunction)
3. p implies q (Implication)

Program (C Code) :

```
1  #include <stdio.h>
2
3  int main() {
4  int p, q;
5  int conjunction, disjunction, implication;
6
7  printf("=====\n");
8  printf(" TRUTH TABLES - LOGICAL EXPRESSIONS \n");
9  printf("=====\n\n");
10
11  /* ---- 1. Conjunction (p AND q) ---- */
12  printf("1. CONJUNCTION (p AND q)\n");
13  printf("-----\n");
14  printf("| p | q | p AND q |\n");
15  printf("-----\n");
16  for (p = 0; p <= 1; p++) {
17  for (q = 0; q <= 1; q++) {
18  conjunction = p && q;
19  printf("| %d | %d | %d |\n", p, q, conjunction);
20  }
21  }
22  printf("-----\n\n");
23
24  /* ---- 2. Disjunction (p OR q) ---- */
25  printf("2. DISJUNCTION (p OR q)\n");
26  printf("-----\n");
27  printf("| p | q | p OR q |\n");
28  printf("-----\n");
29  for (p = 0; p <= 1; p++) {
30  for (q = 0; q <= 1; q++) {
31  disjunction = p || q;
32  printf("| %d | %d | %d |\n", p, q, disjunction);
33  }
```

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34     }
35     printf("-----\\n\\n");
36
37     /* ---- 3. Implication (p IMPLIES q) ---- */
38     /* p -> q is equivalent to (!p) OR q */
39     printf("3. IMPLICATION (p IMPLIES q / p -> q)\\n");
40     printf("-----\\n");
41     printf("| p | q | p -> q |\\n");
42     printf("-----\\n");
43     for (p = 0; p <= 1; p++) {
44         for (q = 0; q <= 1; q++) {
45             implication = (!p) || q;
46             printf("| %d | %d | %d |\\n", p, q, implication);
47         }
48     }
49     printf("-----\\n\\n");
50
51     printf("NOTE: 0 = FALSE, 1 = TRUE\\n");
52     printf("=====\\n");
53
54     return 0;
55 }

```

Output :

```

=====
TRUTH TABLES - LOGICAL EXPRESSIONS
=====

1. CONJUNCTION (p AND q)
-----
| p | q | p AND q |
-----
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |
-----

2. DISJUNCTION (p OR q)
-----

```

p	q	p OR q
---	---	--------

0	0	0
0	1	1
1	0	1
1	1	1

3. IMPLICATION (p IMPLIES q / $p \rightarrow q$)

p	q	$p \rightarrow q$
---	---	-------------------

0	0	1
0	1	1
1	0	0
1	1	1

NOTE: 0 = FALSE, 1 = TRUE

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